

Red Sand (grade 6 edition) — play-through review

Theme `redsand_ms` · middle-school physical science, grade 6 · 10 days, 30 stops · reviewed 2026-08-21 by reading `books/redsand-ms.yml` in full, working every board, and comparing against the parent `redsand`.

Verdict

One of the two best junior editions in the repo, and the reason is structural: `redsand_ms` **was written against** `questionLoad` **from the first stop rather than swept afterwards**, and it shows in every number. `per-day` is `[3,3,3,3,3,3,3,3,3,3]` — thirty stops, ten days, no exceptions, the only junior edition besides `the_trial_ms` and `sightline_ms` with that discipline. It carries **no rows in any debt file**. It computes **5 of 5** syllabus relations. `questionLoad` reports 3 of 30 demanding stops (8%). And it is one of only two junior editions with **nine** — Review **variants**, so it is one of five campaigns in the entire catalogue where a player actually meets an idea a second time — eight of its ten days carry a callback.

The nine boards are correct and the register is right: $340\text{ L} \div 85\text{ L/min} = 4.0$ minutes in the bed; $13.8 \times 44.0 \div 16.0 = 37.9$ kg of carbon dioxide a sol; $\Delta T_f = 3 \times 1.86 \times 8.0 \approx 45\text{ K}$ below pure water's freezing point. The free-energy, reaction-quotient, Faraday and dust-attenuation boards are all present in simplified form.

The parent's plot survives whole, which is the thing worth recording: the window opens on sol 486 and not again for twenty-six months; eleven kilograms a sol against a schedule needing fourteen; Sundqvist wants the reactor hotter and Herrera says a hotter bed hands back less; and the actual fault is a poisoned catalyst nobody was looking at. **Both sides of the argument are named on the opening card with their positions stated**, which eleven campaigns in this catalogue do not manage.

Answerable: 30/30. **Sense:** Excellent, and the ending is the best in the junior set — see below. **Level:** Right, and written to the gate rather than at it. **Fun:** High. "Six people go home on whatever those two can be made to agree."

Findings

ID	Sev	Where	Issue	Proposed fix
RM-01	WORTH	Eight of ten days author 4 calls	<code>per-day</code> counts three <i>authored</i> stops on every day, and the callback makes a fourth on eight of them — which is the day model working exactly as designed and is why this edition has spaced retrieval at all. Recording it because it is the opposite of the finding in the other ten junior editions, where a fourth <i>authored</i> stop displaces the callback.	Nothing. This is the shape . Three authored stops plus a review variant to serve is what <code>shapeMissions</code> was built for, and two junior editions out of twelve do it.
RM-02	WORTH	The parent's RS-01 and RS-02 do not appear here	Worth stating explicitly, because it is a useful control. The parent's equilibrium-constant contradiction ($K = 190$ at 573 K on sol 3 against	Nothing here. When RS-01 is fixed in the parent, check that the fix

K = 0.14 at 593 K on sol 6, with the player's own ΔG implying 3.5×10^5) is **not** in this edition — the junior version simplifies the Q-versus-K comparison and does not quote a second, incompatible K. Nor does the storm ALLOCATE's pool problem appear. Both defects are in the senior book only.

RM-03	WORTH	One stop with no authored background	Thirty-eight guides, thirty-seven backgrounds. One gap.	Close it.
RM-04	TASTE	Numeral damage, 3 occurrences	"1 of", "1 for", "1 hot". The parent has fourteen.	Editorial pass. Cross-campaign §1.
RM-05	TASTE	Two SCIENTETANK stops	Both carry rules, and — unlike the parent's two, which carry neither — worth checking whether they carry evidence. The parent's RS-03 is that its two tank stops let the player allocate a hundred points across four proposals with nothing to argue from.	Check and, if absent, add three or four facts per tank, one per proposal, naming no winner. At grade 6 this matters more than at AP level: a sixth grader with no evidence is guessing, and the format's whole point is that they should not be.

What the derivation did well, and what the set should copy

- **Three authored stops a day, ten days, and nine review variants.** That combination is what makes the day model's spaced retrieval fire. Only `redsand_ms`, `sightline_ms`, `redsand`, `sightline` and `hospital` have it, out of forty-two campaigns.
- **Written to the gate, not swept afterwards.** `MIDDLE_SCHOOL_EDITIONS.md` §9 records this as a different and cheaper process, and the numbers bear it out: 8% demanding stops, no debt rows, 100% of equations computed.
- **Both sides named on the opening card, with positions.** *"Ingrid Sundqvist, who runs production, wants the reactor run hotter tonight. Tomas Herrera, who owns the reactor, says a hotter bed hands back less every pass."*
- **The ending is the best in the junior set:** "the last four hundred kilograms came from a catalyst charge nobody wanted to fit, running on water nobody had been cleaning properly, and both of those were found because somebody kept asking where the missing kilograms had gone." Then the unfinished item is a rule written on paper with no alarm behind it, "and the next crew will find out whether a rule is enough."

Opening and closing

Opening: 112 words, all four beats, both sides named. Model.

Closing: three paragraphs and no weak sentence in any of them. Keep.

Warm-ups

Authored for this edition. No findings.